

BHOOMI · BhuVerify

Intelligent Land Record Digitization & Validation System

Problem Statement ID SIH26018 · Theme: Smart Automation · Category: Software

Converts scanned registers, handwritten pages, PDFs and cadastral maps into structured, validated, GIS-linked digital records — with government reviewers holding final authority.

LEGACY
RECORD

Scanned / Handwritten / PDF /
Image

BHOOMI
INTELLIGENCE

AI OCR · Validation · GIS · Audit

VERIFIED
DIGITAL
RECORD

Structured · Searchable · Auditable

What BHOOMI does · How it transforms legacy land records · Key innovations

SCANNED REGISTER

Odisha 39-A, UP Khatauni, MP Khasra, Bihar Khatiyani

HANDWRITTEN PAGE

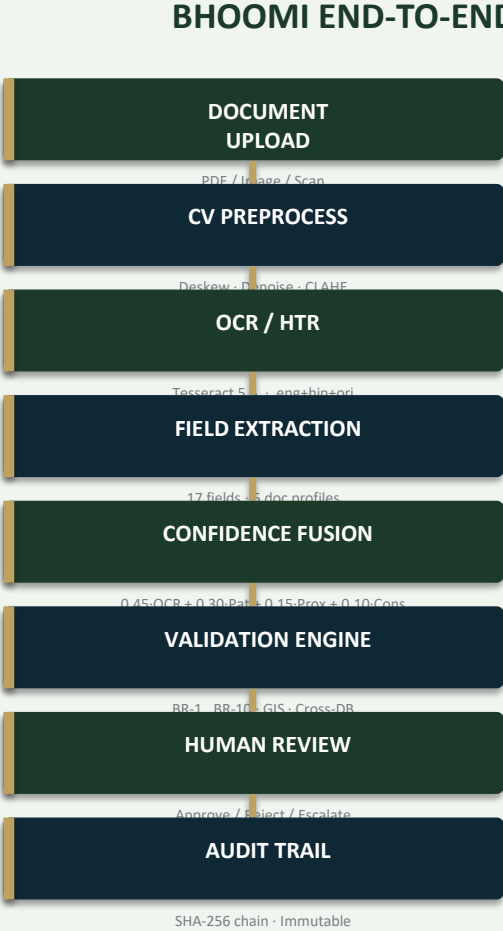
Hindi · Odia · English mixed-script pages

PDF / IMAGE

Multi-page documents, stamps, maps

CADASTRAL MAP

GeoJSON parcel polygons, survey sheets



MULTILINGUAL OCR

English · Hindi · Odia · Script-routed second pass (39-A → Oriya)

FIELD CONFIDENCE

Per-word confidence · Fusion formula · Auto-review routing < 70%

GIS VERIFICATION

Cadastral polygon matching · Area reconciliation · Discrepancy BR-9

HUMAN-IN-THE-LOOP

Government reviewer holds final authority · HTTP 409 blocks approval with open findings



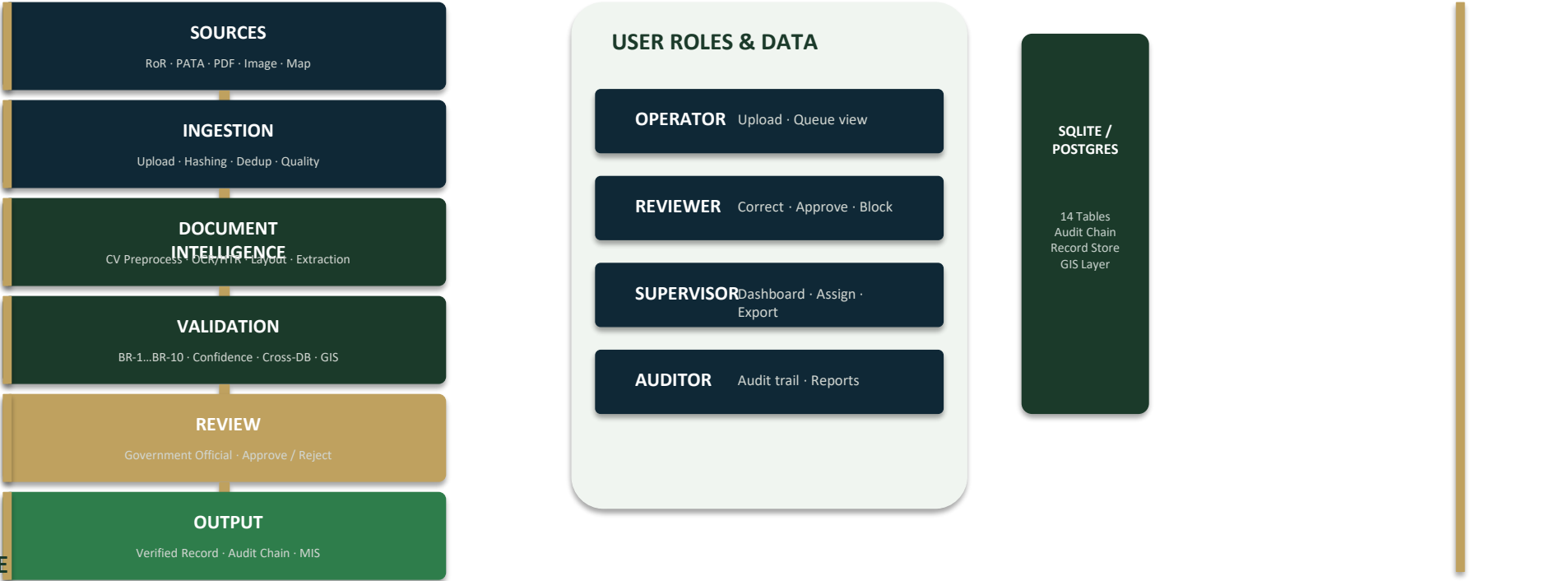
NOT AN OCR WRAPPER — the complete value chain: Document → Intelligence → Validation → GIS → Human Review → Auditable Record

Real document: Odisha RoR · Clean scan

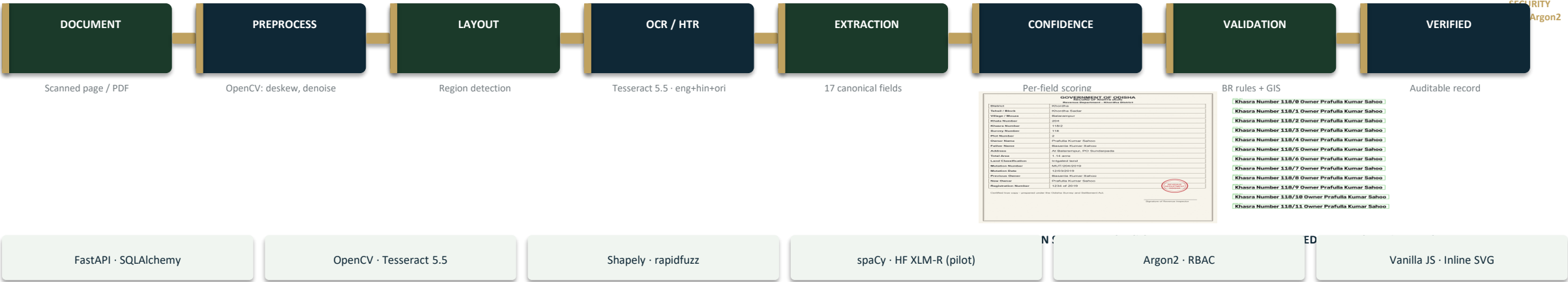
TECHNICAL APPROACH · ARCHITECTURE & PIPELINE

Native architecture diagram · AI pipeline · Real screenshots · Technology stack

BHOOMI SYSTEM ARCHITECTURE



AI DOCUMENT PROCESSING PIPELINE



MEASURED EVIDENCE · PROTOTYPE

165 TESTS

Local / 168 on EC2 · All green

26 DOCUMENTS

6 demo + 5 stress + 5 Hindi + 5 English + 5 real Bhulekh RoRs

79–84% OCR

Real Odisha scans · Multi-script pages

15/15 BYTE-ID

Deterministic reruns · No drift (Groq-off core)

< 1 s/RECORD

Validation latency · 15–127 s/doc (2 vCPU)

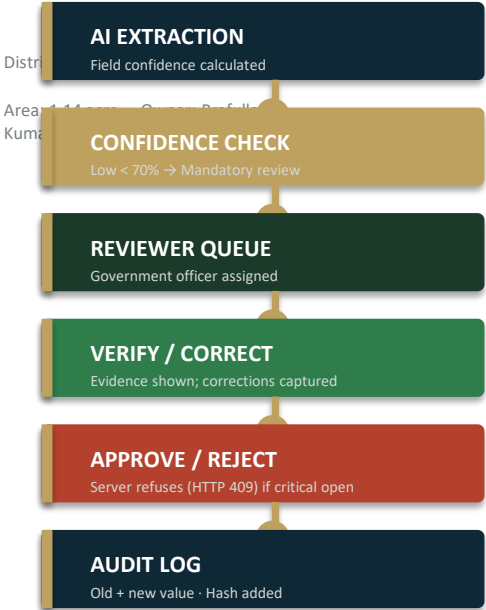
GOVERNMENT OF ODISHA RECORD OF RIGHTS (RoR)	
Revenue Department - Khordha District	
District	Khordha
Tehsil / Block	Khordha Sadar
Village / Mouza	Balarampur
Khata Number	204
Khasra Number	118/2
Survey Number	118
Plot Number	2
Owner Name	Pratula Kumar Sahoo
Father Name	Basanta Kumar Sahoo
Address	At Balarampur, PO Sundarpada
Total Area	1.14 acre
Land Classification	Irrigated land
Mutation Number	MUT/204/2019
Mutation Date	12/03/2019
Previous Owner	Basanta Kumar Sahoo
New Owner	Pratula Kumar Sahoo
Registration Number	1234 of 2019

Certified true copy - prepared under the Odisha Survey and Settlement Act.

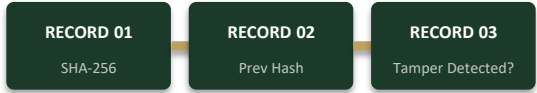
Signature of Revenue Inspector

REAL DOCUMENT · Odisha RoR (Clean)

HUMAN-IN-THE-LOOP WORKFLOW



SECURITY · AUDIT · RISK



Every AI decision and every human correction writes one append-only row. Each row stores SHA-256 of previous row — tampering breaks the chain, detectable via `verify_chain()` (FR-12).

RISK: INVENTED DATA

Anti-hallucination contracts per doc type (e.g., plot 417 is NOT a khasra; khatyan 18 is NOT a khata). AI never writes back without reviewer approval.

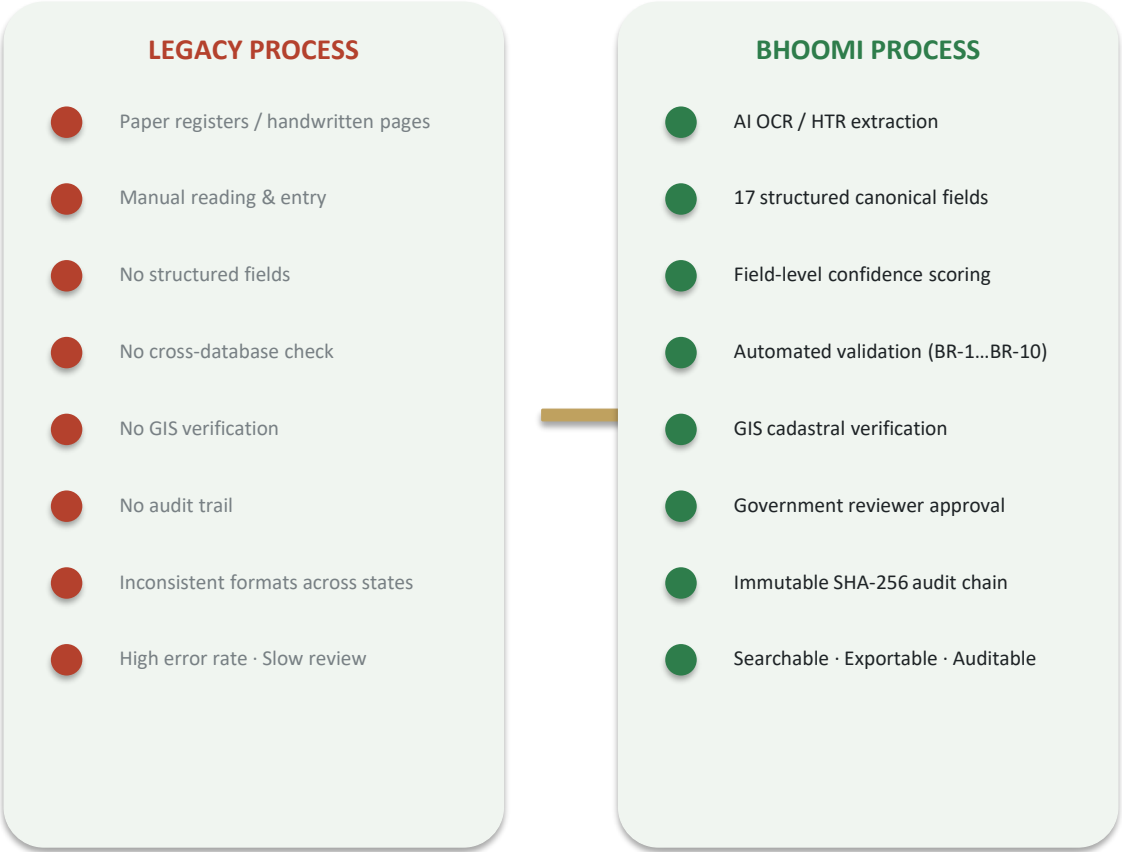
RISK: GARBLAGED OCR

Script-routed second pass + confidence bands. Any result < 70% is routed to mandatory human review. Zero crashes on adversarial set (36.8% accuracy, 100% routed to review).

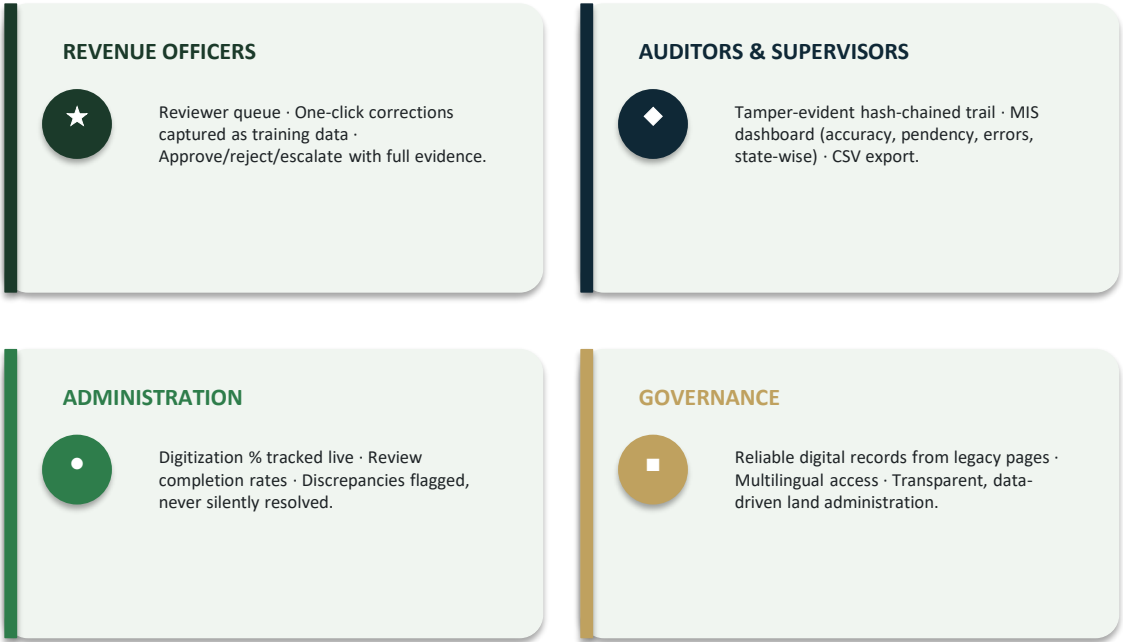
IMPACT & BENEFITS · BEFORE → AFTER · SCALE

Revenue officers · Auditors · Governance · Scalability path · Real impact metrics

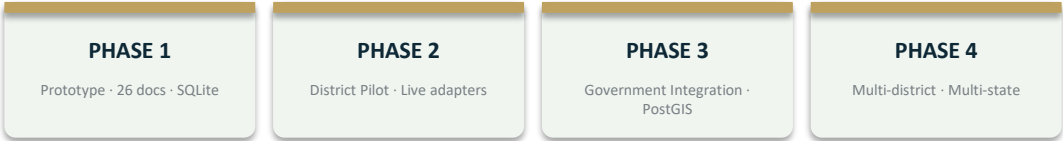
BEFORE → AFTER · THE TRANSFORMATION



IMPACT DIMENSIONS · SCALABILITY



SCALABILITY PATH · PROTOTYPE → PRODUCTION



REFERENCES & SOURCES

SIH PROBLEM STATEMENT

SIH26018 · Ministry of Rural Development · Smart Automation Theme

CODE & REPO

github.com/fresherscomputersc-hash/BHOOMI · FastAPI · SQLAlchemy · Vanilla JS

DOCUMENT INTELLIGENCE

OpenCV · Tesseract 5.5 · spaCy NER · Hugging Face XLM-R · TrOCR (trialed)

GIS & CADASTRAL

Shapely · Sample GeoJSON layer · BhuNaksha integration (adapter-ready)

GOVERNMENT SYSTEMS

Bhulekh / RoR · BhuNaksha · IGR · LGD · DILRMP / LRMS — mock adapters, live-ready contracts

SECURITY & AUDIT

SHA-256 chain · Argon2 hashing · RBAC · Audit trail with re-verify

GOVERNMENT ADAPTER LAYER

BHOOMI ADAPTER LAYER

Live-ready contracts · Mock data now

BHULEKH / RoR

Ownership verification

MOCK · LIVE
CONTRACT READY

BHU NAKSHA

Cadastral map reference

MOCK · LIVE
CONTRACT READY

IGR

Registration / deed

MOCK · LIVE
CONTRACT READY

LRMS / DILRMP

Downstream status

MOCK · LIVE
CONTRACT READY

LGD MASTER

Village / tehsil check

MOCK · LIVE
CONTRACT READY

PRODUCTION SWAP · ROADMAP

DATABASE

SQLite → PostgreSQL + PostGIS

QUEUE

In-process → Redis / Celery

HTR ENGINE

Tesseract → TrOCR / IndicHTR

LAYOUT

OpenCV heuristics → YOLO / Detectron2

AUTH

Seeded + Argon2 → Department SSO + MFA

GIS SERVING

Sample GeoJSON → GeoServer

ROADMAP

1

PHASE 1

Prototype · 26 docs

2

PHASE 2

District Pilot · Live adapters

3

PHASE 3

Govt Integration · PostGIS

4

PHASE 4

Multi-state Scale